barilla-guoth-pseja

fguot

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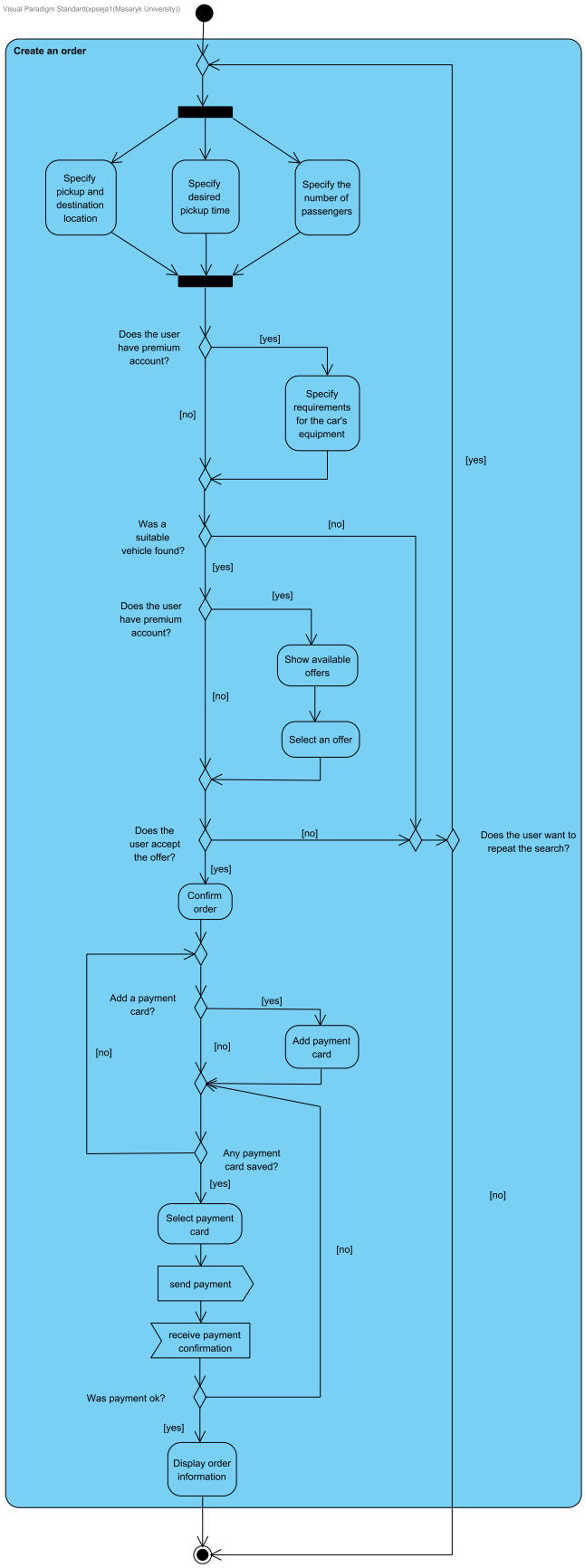
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Activity Diagram

Create an order Activity Diagram



Summary

Name	Description
Create an order	
Specify pickup and destination location	
Specify desired pickup time	
Specify the number of passengers	
Does the user have premium account?	
Specify requirements for the car's equipment	
Was a suitable vehicle found?	
Does the user have premium account?	
Show available offers	
Select an offer	
Does the user want to repeat the search?	
Does the user accept the offer?	
Confirm order	
Add a payment card?	
Add payment card	
Any payment card saved?	
Select payment card	
send payment	
receive payment confirmation	
Was payment ok?	
Display order information	

Details

N/A

Create an order

Children

Name	Description
Does the user have premium account?	
N/A	
N/A	
Specify pickup and destination location	
Specify desired pickup time	
Specify the number of passengers	
Select payment card	
Specify requirements for the car's equipment	
N/A	
N/A	
Was a suitable vehicle found?	
Confirm order	
Does the user have premium account?	
Does the user accept the offer?	
Select an offer	
Add a payment card?	
N/A	
Any payment card saved?	
Add payment card	
N/A	
send payment	
receive payment confirmation	
N/A	
Show available offers	
Does the user want to repeat the search?	
Display order information	
Was payment ok?	
N/A	

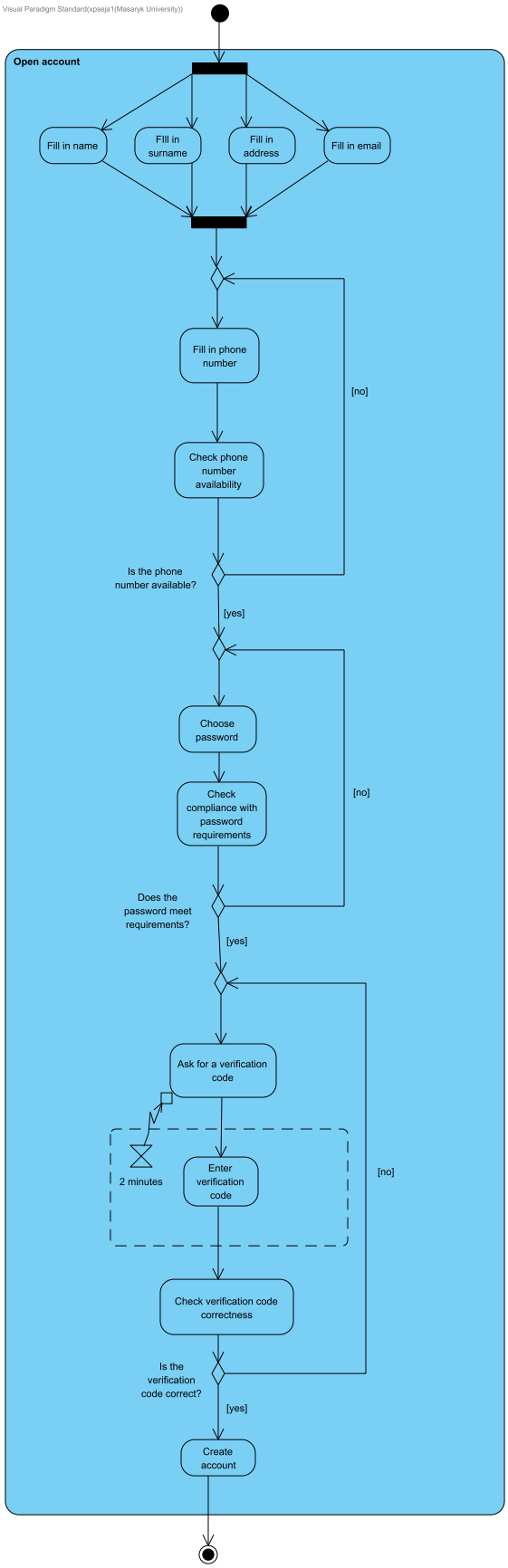
N/A
N/A
Specify pickup and destination location
Specify desired pickup time
Specify the number of passengers
N/A
Does the user have premium account?

Specify requirements for the car's equipment
N/A
Was a suitable vehicle found?
Does the user have premium account?
Show available offers

Select an offer
N/A
Does the user want to repeat the search?
N/A
N/A
Does the user accept the offer?
Confirm order
N/A
Add a payment card?
Add payment card
N/A
Any payment card saved?
Select payment card
send payment
receive payment confirmation
Was payment ok?
Display order information

Activity Diagram

Open account Activity Diagram



Summary

Name	Description
Open account	
Fill in name	
FIll in surname	
Fill in address	
Fill in email	
Fill in phone number	
Check phone number availability	
Is the phone number available?	
Choose password	
Check compliance with password requirements	
Does the password meet requirements?	
Ask for a verification code	
Enter verification code	
Region	
Check verification code correctness	
Is the verification code correct?	
Create account	
2 minutes	

Details

N/A

N/A

Open account

Children

Name	Description
Region	
N/A	
Fill in name	
Fill in address	
FIll in surname	
Fill in email	
N/A	
Fill in phone number	
N/A	
Is the phone number available?	
Choose password	
Does the password meet requirements?	
N/A	
N/A	
Ask for a verification code	
Is the verification code correct?	
Create account	
Check phone number availability	
Check compliance with password requirements	
Check verification code correctness	

Fill in name
FIll in surname
Fill in address
Fill in email
N/A
N/A
Fill in phone number
Check phone number availability
Is the phone number available?
N/A
Choose password

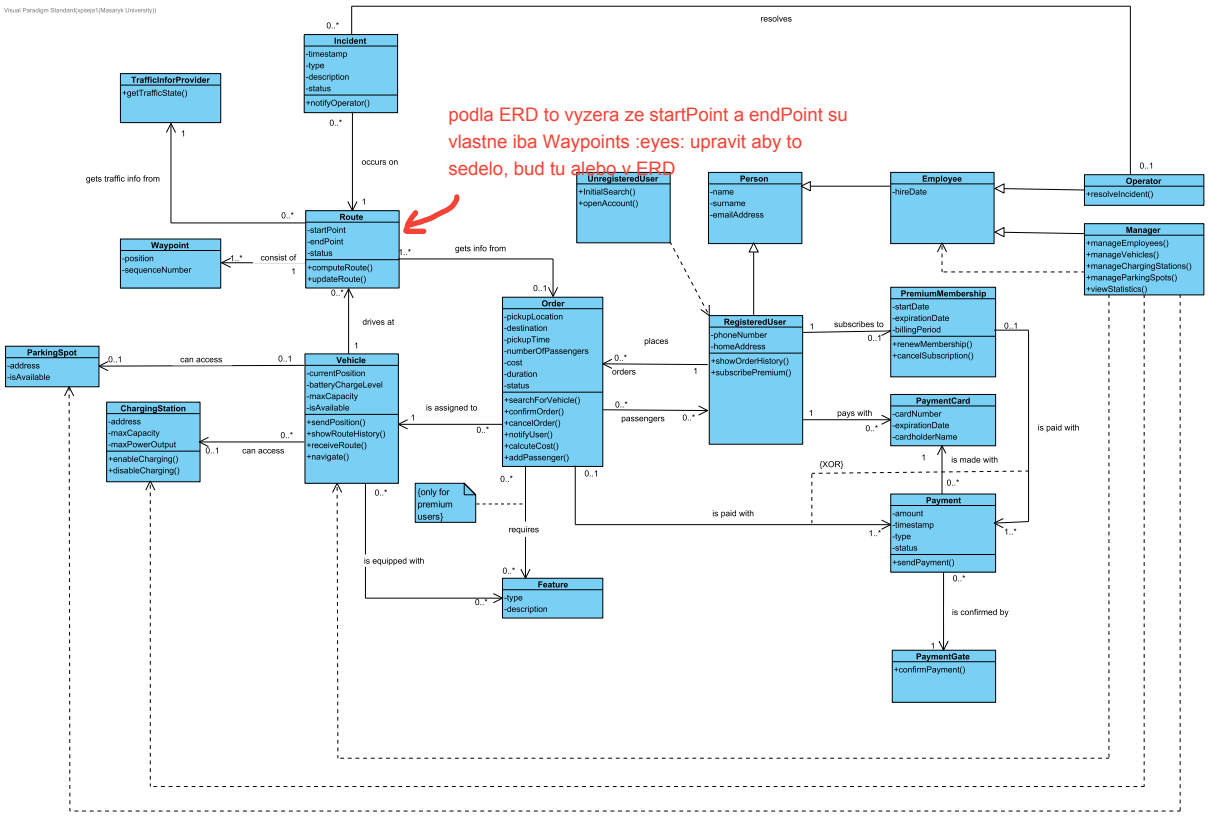
Check compliance with password requirements
Does the password meet requirements?
N/A
N/A
Ask for a verification code
Enter verification code
Region
Children

Name	Description
2 minutes	
Enter verification code	

Check verification code correctness
Is the verification code correct?
Create account
N/A
2 minutes

Class Diagram

SmartTaxIS - Class Diagram



Payment
- type: RIDE / CANCELLATION_FEE / SUBSCRIPTION
- status: PENDING / COMPLETED / FAILED
- exactly one of Order or PremiumMembership

Order
- status: CONFIRMED / IN_PROGRESS / COMPLETED / CANCELLED

Incident
- status: OPEN / RESOLVED

PremiumMembership
- billingPeriod: MONTHLY / ANNUAL
- renewed automatically, notification 3 days before expiration

Employee
- Operator: notified immediately when an incident occurs

Summary

Name	Description
Incident	
TrafficInforProvider	
UnregisteredUser	
Person	

Employee	
Operator	
Route	
Manager	
Waypoint	
PremiumMembership	
Order	
RegisteredUser	
ParkingSpot	
Vehicle	
PaymentCard	
ChargingStation	
Payment	
Feature	
PaymentGate	
N/A	{only for premium users}
N/A	Payment - type: RIDE / CANCELLATION_FEE / SUBSCRIPTION - status: PENDING / COMPLETED / FAILED - exactly one of Order or PremiumMembership Order - status: CONFIRMED / IN_PROGRESS / COMPLETED / CANCELLED Incident - status: OPEN / RESOLVED PremiumMembership - billingPeriod: MONTHLY / ANNUAL - renewed automatically, notification 3 days before expiration Employee - Operator: notified immediately when an incident occurs - Manager: manages employees, vehicles, charging stations, parking spots

Details

Incident

TrafficInforProvider

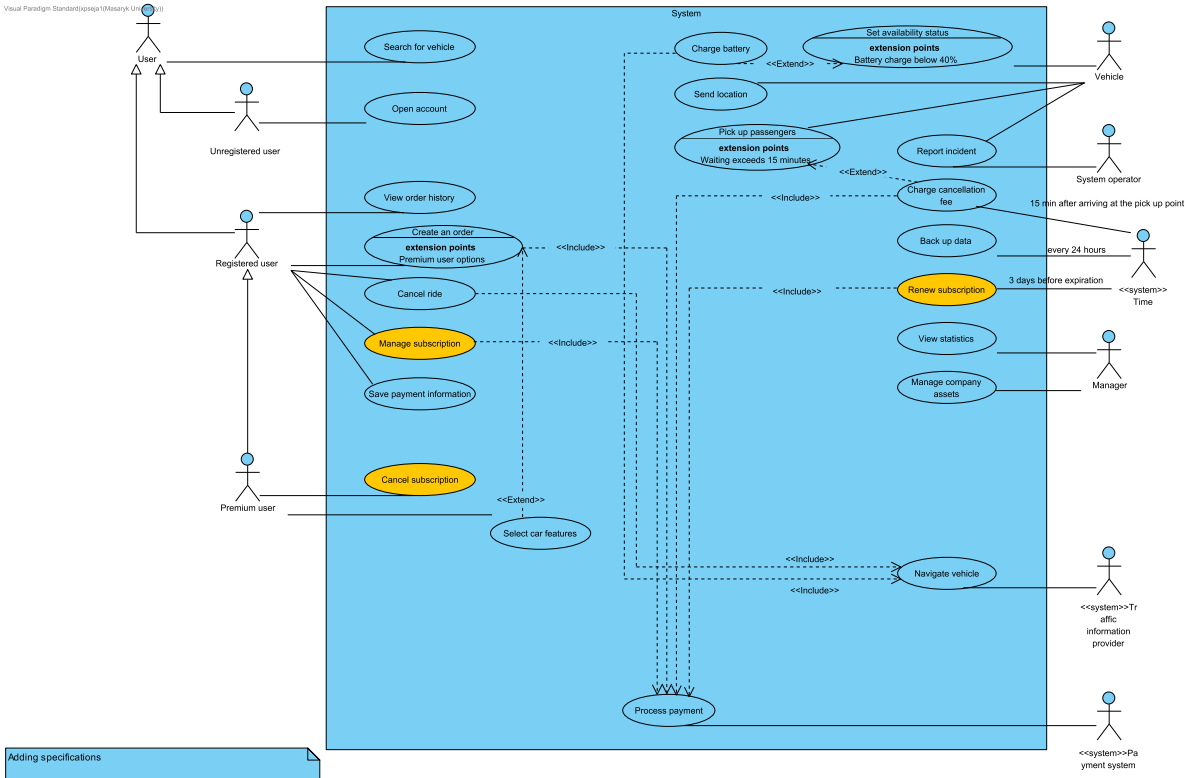
UnregisteredUser

Person

Employee
Operator
Route
Manager
Waypoint
PremiumMembership
Order
RegisteredUser
ParkingSpot
Vehicle
PaymentCard
ChargingStation
Payment
Feature
PaymentGate
N/A
N/A

Use Case Diagram

SmartTaxIS – Use Case Diagram



Adding specifications

Q: How much is the cancellation fee?
Assumption: A fixed fee of 5 € is charged.

Q: How does the user add additional passengers to the order?
Assumption: By entering their registered e-mail addresses or username.

Q: How many offers are displayed to a premium user? Assumption: Up to 3 offers, sorted by the selection criteria.

Q: What happens if all charging stations are fully occupied?
Assumption: The vehicle is navigated to the nearest station and waits for a free slot.

Data backup is not mentioned in the specification. Agreed: the system backs up all data every day at 23:59.

Summary

Name	Description
User	
Unregistered user	
Registered user	
Premium user	
Vehicle	
System operator	
Manager	
<<system>>Traffic information provider	
<<system>>Payment system	
<<system>>Time	
Report incident	

Use Case ID	UC38
Brief Description	If a problem occurs during a route (e.g. an accident or the vehicle cannot continue), the vehicle reports it to the system. The system immediately notifies the system operator, who takes care of the incident. The incident is stored in the vehicle's history together with its routes and route changes for troubleshooting purposes.
Primary Actors	Vehicle
Secondary Actors	System operator
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

Renew subscription

Use Case ID	UC37
Brief Description	Three days before the end of the current billing period, the system notifies the premium user about the upcoming renewal. At the end of the period, the subscription is renewed automatically and the user is charged through the external payment system.
Primary Actors	Time
Secondary Actors	Payment system
Preconditions	1. The user has an active premium subscription. 2. The subscription has not been cancelled.
Main Flow of Events	1. Three days before the end of the current billing period, the system sends a notification to the premium user about the upcoming renewal and its price. 2. The current billing period ends.

		3. The system checks that the subscription has not been cancelled. 4. INCLUDE: Process payment - the system charges the user for the next billing period. 5. The system extends the subscription by one billing period (month or year). 6. The system informs the user that the subscription has been renewed.
	Alternative Flows	3a. The subscription was cancelled 1. The system does not renew the subscription. 2. The system deactivates the premium membership and the user becomes a registered user again. 3. The use case ends. 4a. The payment fails 1. The system notifies the user that the payment was not successful. 2. The system deactivates the premium membership. 3. The use case ends.
	Post-conditions	1. The subscription is extended for another billing period, or 2. The premium membership is deactivated.

Pick up passengers

Use Case ID	UC36
Brief Description	When the vehicle arrives at the pickup location, the system notifies the user. The vehicle waits until all passengers get on board and then heads to the destination.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	

	Post-conditions	
Navigate vehicle		
	Use Case ID	UC35
	Brief Description	The system computes a route as a collection of waypoints based on the current position of the vehicle and the traffic situation obtained from the external traffic information provider. The route is sent to the vehicle, which follows it autonomously.
	Primary Actors	
	Secondary Actors	
	Preconditions	
	Main Flow of Events	
	Alternative Flows	
	Post-conditions	
Cancel ride		
	Use Case ID	UC34
	Brief Description	A passenger can request cancellation of the ride at any time. The system finds a suitable place to park, updates the vehicle's destination and sends the new route to the vehicle.
	Primary Actors	
	Secondary Actors	
	Preconditions	
	Main Flow of Events	

	Alternative Flows	
	Post-conditions	
N/A	Adding specifications Q: How much is the cancellation fee? Assumption: A fixed fee of 5 € is charged. Q: How does the user add additional passengers to the order? Assumption: By entering their registered e-mail addresses or username. Q: How many offers are displayed to a premium user? Assumption: Up to 3 offers, sorted by the selection criteria. Q: What happens if all charging stations are fully occupied? Assumption: The vehicle is navigated to the nearest station and waits for a free slot. Data backup is not mentioned in the specification. Agreed: the system backs up all data every day at 23:59.
Charge cancellation fee	
	Use Case ID	UC29
	Brief Description	When the vehicle arrives at the pickup location, it waits until all passengers get on board. If the waiting time exceeds 15 minutes, the system automatically cancels the order and charges the user a cancellation fee through the external payment system.
	Primary Actors	
	Secondary Actors	
	Preconditions	
	Main Flow of Events	

	Alternative Flows
	Post-conditions
Back up data	
	Use Case ID UC28
	Brief Description System backs up all data at 23:59 everyday.
	Primary Actors
	Secondary Actors
	Preconditions
	Main Flow of Events
	Alternative Flows
	Post-conditions
Process payment	
	Use Case ID UC27
	Brief Description This use case allows the system to charge the user for a service through an external online payment provider.
	Primary Actors
	Secondary Actors
	Preconditions
	Main Flow of Events
	Alternative Flows
	Post-conditions
Cancel subscription	

	Use Case ID	UC26
	Brief Description	A premium user can cancel the subscription at any time. The premium benefits remain active until the end of the current billing period, after which the subscription is not renewed and the user becomes a registered user again.
	Primary Actors	Premium user
	Secondary Actors	-
	Preconditions	1. The user is logged in to the system. 2. The user has an active premium subscription.
	Main Flow of Events	1. The premium user selects the option to cancel the subscription. 2. The system displays the end date of the current billing period and informs the user that the premium benefits remain active until then. 3. The system asks the user to confirm the cancellation. 4. The premium user confirms the cancellation. 5. The system marks the subscription as cancelled so that it will not be renewed. 6. The system confirms the cancellation to the user.
	Alternative Flows	4a. The user does not confirm the cancellation 1. The premium user leaves the cancellation dialog. 2. The subscription remains active and will be renewed as usual.
	Post-conditions	1. The subscription is marked as cancelled. 2. The user keeps the premium benefits until the end of the current billing period, then loses premium status (Renew subscription, alternative flow 3a).
Set availability status		
	Use Case ID	UC25
	Brief Description	Vehicle sets its availability status in the system as long as it is parked in the

		designated place and its battery charge is above 40%.
	Primary Actors	Vehicle
	Secondary Actors	-
	Preconditions	1. Vehicle is parked in the designated place 2. Vehicle's battery charge is above 40%
	Main Flow of Events	1. After last action (charging/order) vehicle relocates to a suitable place to park 2. Vehicle updates its position 3. Vehicle checks the battery charge level
	Alternative Flows	Battery charge level below 40% 1. Vehicle is navigated to the nearest free charging station 2. Charging is disabled once the battery charge exceeds 90% 3. Vehicle is navigated to a suitable parking space
	Post-conditions	
Send location		
	Use Case ID	UC23
	Brief Description	The vehicle sends its position to the system every second.
	Primary Actors	
	Secondary Actors	
	Preconditions	
	Main Flow of Events	
	Alternative Flows	
	Post-conditions	
Search for vehicle		
	Use Case ID	UC22

Brief Description	User can search through available vehicles for a ride.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

View statistics

Use Case ID	UC21
Brief Description	Manager can view the statistics of the company.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

Manage company assets

Use Case ID	UC14
Brief Description	The manager can manage all assets associated with the taxi company, i.e. create, update and delete their records. This is a general use case specialized by Manage employees, Manage vehicles,

		Manage charging stations and Manage parking spots.
	Primary Actors	
	Secondary Actors	
	Preconditions	
	Main Flow of Events	
	Alternative Flows	
	Post-conditions	

Charge battery		
	Use Case ID	UC12
	Brief Description	When vehicle's battery charge level drops below 40%, it is navigated to the nearest free charging station.
	Primary Actors	
	Secondary Actors	
	Preconditions	
	Main Flow of Events	
	Alternative Flows	
	Post-conditions	

Select car features		
	Use Case ID	UC10
	Brief Description	Premium user can select from available car features.
	Primary Actors	

	Secondary Actors	
	Preconditions	
	Main Flow of Events	
	Alternative Flows	
	Post-conditions	
Manage subscription		
	Use Case ID	UC08
	Brief Description	A registered user can subscribe to a premium membership and become a premium user. The user chooses whether the subscription is billed monthly or annually and pays the first period through the external payment system. Renewal of the subscription is handled by Renew subscription.
	Primary Actors	Registered user
	Secondary Actors	Payment system
	Preconditions	1. The user is registered and logged in to the system. 2. The user does not have an active premium subscription.
	Main Flow of Events	1. The registered user selects the option to purchase a premium membership. 2. The system displays the premium benefits (broader search options, 20 % discount on all orders) and the available billing options (monthly / annually) with their prices. 3. The registered user selects the billing option. 4. The system displays a summary of the subscription and asks the user to confirm it. 5. The registered user confirms the subscription.

		6. INCLUDE: Process payment - the system charges the user for the first billing period using the stored payment card. 7. The system activates the premium membership and sets the end date of the current billing period. 8. The system informs the user that the subscription is active and will be renewed automatically.
	Alternative Flows	3a. The user cancels the purchase 1. The user leaves the purchase before confirming it. 2. The use case ends without any changes. 6a. The user has no stored payment card 1. The system asks the user to enter payment card details. 2. The user enters the details, which are saved in the system. 3. The use case continues at step 6. 6b. The payment fails 1. The system informs the user that the payment was not successful. 2. The premium membership is not activated and the use case ends.
	Post-conditions	1. The user is a premium user and can use the premium benefits. 2. The subscription will be renewed automatically (Renew subscription).
View order history		
	Use Case ID	UC04
	Brief Description	Registered user can view his order history.
	Primary Actors	
	Secondary Actors	
	Preconditions	
	Main Flow of Events	

Alternative Flows	
Post-conditions	

Save payment information

Use Case ID	UC03
Brief Description	Registered user can save his payment information for the next payment.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	
Post-conditions	

Create an order

Use Case ID	UC02
Brief Description	Registered user can create an order. Registered user confirms one of the offers found by the vehicle search. Unregistered users can search for vehicles but must create an account before confirming an order.
Primary Actors	
Secondary Actors	
Preconditions	
Main Flow of Events	
Alternative Flows	

	Post-conditions	
Open account		
	Use Case ID	UC01
	Brief Description	Unregistered user can create a new account.
	Primary Actors	
	Secondary Actors	
	Preconditions	
	Main Flow of Events	
	Alternative Flows	
	Post-conditions	
System		

Description

Functional Requirements

User
FR1. The system allows searching for a vehicle.

Unregistered User
FR2. The system allows the creation of an account.

Registered User
FR3. The system allows the creation of an order.
FR4. The system allows browsing the order history.
FR5. The system allows users to save their payment details.
FR6. The system allows the purchase of a premium membership.
FR7. The system allows a passenger to cancel the ride.

Premium User
FR8. The system allows the selection of vehicle equipment.
FR9. The system allows cancelling the subscription.

Vehicle

FR10. The system tracks the current location of the vehicle.
FR11. The system monitors the availability status of the vehicle.
FR12. The system computes routes and navigates the vehicle.
FR13. The system notifies the user when the vehicle arrives at the pickup location.

Manager
FR14. The system allows the manager to manage the company's assets.
FR15. The system displays statistics.

System operator
FR16. The system notifies the system operator about incidents during a route.

Time
FR17. The system charges a fee for order cancellation.
FR18. The system backs up data every day at 23:59.
FR19. The system automatically renews subscriptions.

Payment system
FR20. The external payment system executes payments.

Non-Functional Requirements
NFR1. The vehicles managed by the system are electric and fully autonomous.
NFR2. Each vehicle sends its position to the system at least once per second.
NFR3. The system must be available 24/7.
NFR4. The system must display vehicle search results within three seconds.
NFR5. Stored payment card information must be encrypted and handled securely.

Details

User
Unregistered user
Registered user
Premium user
Vehicle
System operator
Manager
<<system>>Traffic information provider
<<system>>Payment system
<<system>>Time
Report incident

Renew subscription

Pick up passengers

Extension Points

Waiting exceeds 15 minutes	
Backlog Activity Id	0

Navigate vehicle

Cancel ride

N/A

Charge cancellation fee

Back up data

Process payment

Cancel subscription

Set availability status

Extension Points

Battery charge below 40%	
Backlog Activity Id	0

Send location

Search for vehicle

View statistics

Manage company assets

Charge battery

Select car features

Manage subscription

View order history

Save payment information

Create an order

Extension Points

Premium user options	
Backlog Activity Id	0

Sub Diagrams

Name	Description
Create an order Activity Diagram	

Open account

Sub Diagrams

Name	Description
Open account Activity Diagram	

System

Children

Name	Description
Open account	<table><tr><td>Use Case ID</td><td>UC01</td></tr><tr><td>Brief Description</td><td>Unregistered user can create a new account.</td></tr><tr><td>Primary Actors</td><td></td></tr><tr><td>Secondary Actors</td><td></td></tr><tr><td>Preconditions</td><td></td></tr><tr><td>Main Flow of Events</td><td></td></tr><tr><td>Alternative Flows</td><td></td></tr><tr><td>Post-conditions</td><td></td></tr></table>
Create an order	<table><tr><td>Use Case ID</td><td>UC02</td></tr><tr><td>Brief Description</td><td>Registered user can create an order. Registered user confirms one of the offers found by the vehicle search. Unregistered users can search for vehicles but must create an account before confirming an order.</td></tr><tr><td>Primary Actors</td><td></td></tr><tr><td>Secondary Actors</td><td></td></tr><tr><td>Preconditions</td><td></td></tr></table>

	Main Flow of Events	
	Alternative Flows	
	Post-conditions	

View order history		
	Use Case ID	UC04
	Brief Description	Registered user can view his order history.
	Primary Actors	
	Secondary Actors	
	Preconditions	
	Main Flow of Events	
	Alternative Flows	
	Post-conditions	

Save payment information		
	Use Case ID	UC03
	Brief Description	Registered user can save his payment information for the next payment.
	Primary Actors	
	Secondary Actors	
	Preconditions	
	Main Flow of Events	
	Alternative Flows	

	Post-conditions	
Manage subscription		
	Use Case ID	UC08
	Brief Description	A registered user can subscribe to a premium membership and become a premium user. The user chooses whether the subscription is billed monthly or annually and pays the first period through the external payment system. Renewal of the subscription is handled by Renew subscription.
	Primary Actors	Registered user
	Secondary Actors	Payment system
	Preconditions	1. The user is registered and logged in to the system. 2. The user does not have an active premium subscription.
	Main Flow of Events	1. The registered user selects the option to purchase a premium membership. 2. The system displays the premium benefits (broader search options, 20 % discount on all orders) and the available billing options (monthly / annually) with their prices. 3. The registered user selects the billing option. 4. The system displays a summary of the subscription and asks the user to confirm it. 5. The registered user confirms the subscription. 6. INCLUDE: Process payment - the system charges the user for the first billing period using the stored payment card. 7. The system activates the premium membership and sets the end date of the current billing period. 8. The system informs the user that the subscription is active and will be renewed automatically.
	Alternative Flows	3a. The user cancels the purchase 1. The user leaves the purchase before confirming it. 2. The use case ends without any changes.

		6a. The user has no stored payment card 1. The system asks the user to enter payment card details. 2. The user enters the details, which are saved in the system. 3. The use case continues at step 6. 6b. The payment fails 1. The system informs the user that the payment was not successful. 2. The premium membership is not activated and the use case ends.
	Post-conditions	1. The user is a premium user and can use the premium benefits. 2. The subscription will be renewed automatically (Renew subscription).
Select car features		
	Use Case ID	UC10
	Brief Description	Premium user can select from available car features.
	Primary Actors	
	Secondary Actors	
	Preconditions	
	Main Flow of Events	
	Alternative Flows	
	Post-conditions	
Charge battery		
	Use Case ID	UC12
	Brief Description	When vehicle's battery charge level drops below 40%, it is navigated to the nearest free charging station.
	Primary Actors	

	Secondary Actors	
	Preconditions	
	Main Flow of Events	
	Alternative Flows	
	Post-conditions	
Manage company assets		
	Use Case ID	UC14
	Brief Description	The manager can manage all assets associated with the taxi company, i.e. create, update and delete their records. This is a general use case specialized by Manage employees, Manage vehicles, Manage charging stations and Manage parking spots.
	Primary Actors	
	Secondary Actors	
	Preconditions	
	Main Flow of Events	
	Alternative Flows	
	Post-conditions	
View statistics		
	Use Case ID	UC21
	Brief Description	Manager can view the statistics of the company.
	Primary Actors	
	Secondary Actors	

	Preconditions	
	Main Flow of Events	
	Alternative Flows	
	Post-conditions	

Search for vehicle	
	Use Case ID	UC22
	Brief Description	User can search through available vehicles for a ride.
	Primary Actors	
	Secondary Actors	
	Preconditions	
	Main Flow of Events	
	Alternative Flows	
	Post-conditions	

Send location	
	Use Case ID	UC23
	Brief Description	The vehicle sends its position to the system every second.
	Primary Actors	
	Secondary Actors	
	Preconditions	
	Main Flow of Events	

	Alternative Flows	
	Post-conditions	
Set availability status		
	Use Case ID	UC25
	Brief Description	Vehicle sets its availability status in the system as long as it is parked in the designated place and its battery charge is above 40%.
	Primary Actors	Vehicle
	Secondary Actors	
	Preconditions	1. Vehicle is parked in the designated place 2. Vehicle's battery charge is above 40%
	Main Flow of Events	1. After last action (charging/order) vehicle relocates to a suitable place to park 2. Vehicle updates its position 3. Vehicle checks the battery charge level
	Alternative Flows	Battery charge level below 40% 1. Vehicle is navigated to the nearest free charging station 2. Charging is disabled once the battery charge exceeds 90% 3. Vehicle is navigated to a suitable parking space
	Post-conditions	
Cancel subscription		
	Use Case ID	UC26
	Brief Description	A premium user can cancel the subscription at any time. The premium benefits remain active until the end of the current billing period, after which the subscription is not renewed and the user becomes a registered user again.
	Primary Actors	Premium user
	Secondary Actors	

	Preconditions	1. The user is logged in to the system. 2. The user has an active premium subscription.
	Main Flow of Events	1. The premium user selects the option to cancel the subscription. 2. The system displays the end date of the current billing period and informs the user that the premium benefits remain active until then. 3. The system asks the user to confirm the cancellation. 4. The premium user confirms the cancellation. 5. The system marks the subscription as cancelled so that it will not be renewed. 6. The system confirms the cancellation to the user.
	Alternative Flows	4a. The user does not confirm the cancellation 1. The premium user leaves the cancellation dialog. 2. The subscription remains active and will be renewed as usual.
	Post-conditions	1. The subscription is marked as cancelled. 2. The user keeps the premium benefits until the end of the current billing period, then loses premium status (Renew subscription, alternative flow 3a).
Process payment		
	Use Case ID	UC27
	Brief Description	This use case allows the system to charge the user for a service through an external online payment provider.
	Primary Actors	
	Secondary Actors	
	Preconditions	
	Main Flow of Events	
	Alternative Flows	
	Post-conditions	

<table>
<tr><td>Back up data</td><td><table>
<tr><td>Use Case ID</td><td>UC28</td></tr>
<tr><td>Brief Description</td><td>System backs up all data at 23:59 everyday.</td></tr>
<tr><td>Primary Actors</td><td></td></tr>
<tr><td>Secondary Actors</td><td></td></tr>
<tr><td>Preconditions</td><td></td></tr>
<tr><td>Main Flow of Events</td><td></td></tr>
<tr><td>Alternative Flows</td><td></td></tr>
<tr><td>Post-conditions</td><td></td></tr>
</table></td></tr>
<tr><td>Charge cancellation fee</td><td><table>
<tr><td>Use Case ID</td><td>UC29</td></tr>
<tr><td>Brief Description</td><td>When the vehicle arrives at the pickup location, it waits until all passengers get on board. If the waiting time exceeds 15 minutes, the system automatically cancels the order and charges the user a cancellation fee through the external payment system.</td></tr>
<tr><td>Primary Actors</td><td></td></tr>
<tr><td>Secondary Actors</td><td></td></tr>
<tr><td>Preconditions</td><td></td></tr>
<tr><td>Main Flow of Events</td><td></td></tr>
<tr><td>Alternative Flows</td><td></td></tr>
<tr><td>Post-conditions</td><td></td></tr>
</table></td></tr>
<tr><td>Cancel ride</td><td><table>
<tr><td>Use Case ID</td><td>UC34</td></tr>
</table></td></tr>
</table>

	Brief Description	A passenger can request cancellation of the ride at any time. The system finds a suitable place to park, updates the vehicle's destination and sends the new route to the vehicle.
	Primary Actors	
	Secondary Actors	
	Preconditions	
	Main Flow of Events	
	Alternative Flows	
	Post-conditions	
Pick up passengers		
	Use Case ID	UC36
	Brief Description	When the vehicle arrives at the pickup location, the system notifies the user. The vehicle waits until all passengers get on board and then heads to the destination.
	Primary Actors	
	Secondary Actors	
	Preconditions	
	Main Flow of Events	
	Alternative Flows	
	Post-conditions	
Navigate vehicle		
	Use Case ID	UC35

	Brief Description	The system computes a route as a collection of waypoints based on the current position of the vehicle and the traffic situation obtained from the external traffic information provider. The route is sent to the vehicle, which follows it autonomously.
	Primary Actors	
	Secondary Actors	
	Preconditions	
	Main Flow of Events	
	Alternative Flows	
	Post-conditions	
Renew subscription		
	Use Case ID	UC37
	Brief Description	Three days before the end of the current billing period, the system notifies the premium user about the upcoming renewal. At the end of the period, the subscription is renewed automatically and the user is charged through the external payment system.
	Primary Actors	Time
	Secondary Actors	Payment system
	Preconditions	1. The user has an active premium subscription. 2. The subscription has not been cancelled.
	Main Flow of Events	1. Three days before the end of the current billing period, the system sends a notification to the premium user about the upcoming renewal and its price. 2. The current billing period ends. 3. The system checks that the subscription has not been cancelled. 4. INCLUDE: Process payment - the system charges the user for the next billing period.

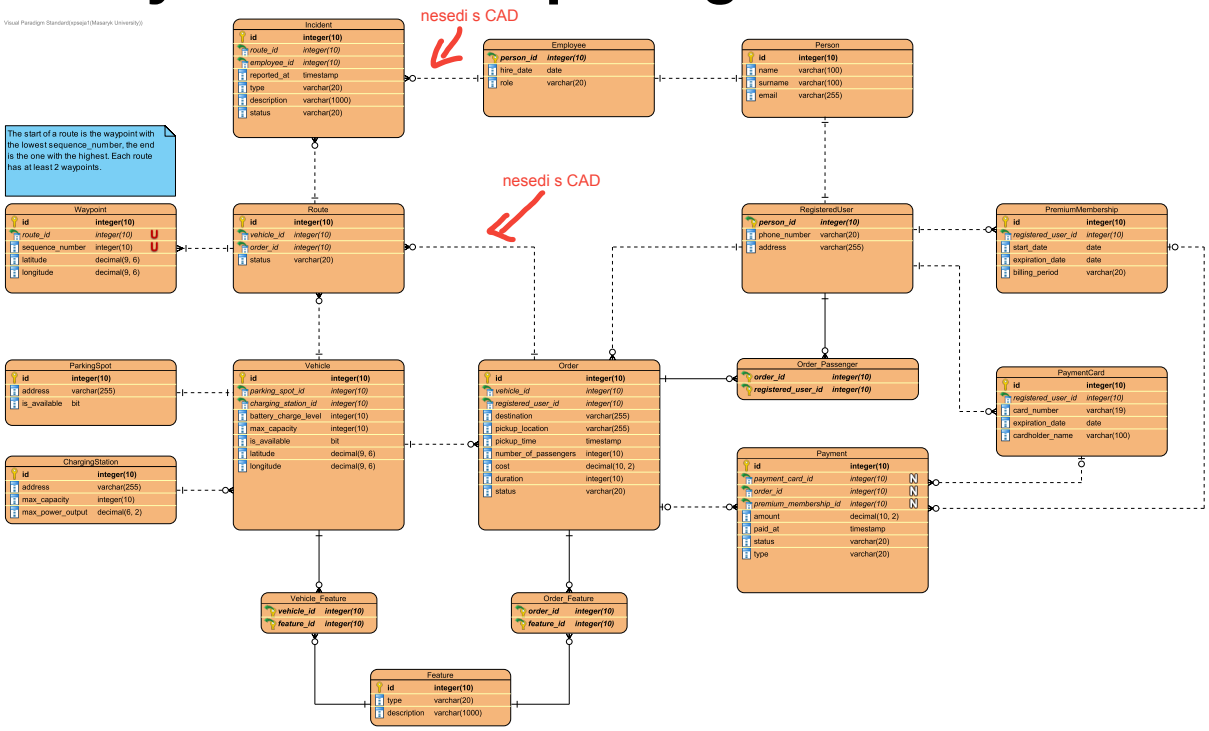
		5. The system extends the subscription by one billing period (month or year). 6. The system informs the user that the subscription has been renewed.
	Alternative Flows	3a. The subscription was cancelled 1. The system does not renew the subscription. 2. The system deactivates the premium membership and the user becomes a registered user again. 3. The use case ends. 4a. The payment fails 1. The system notifies the user that the payment was not successful. 2. The system deactivates the premium membership. 3. The use case ends.
	Post-conditions	1. The subscription is extended for another billing period, or 2. The premium membership is deactivated.
Report incident		
	Use Case ID	UC38
	Brief Description	If a problem occurs during a route (e.g. an accident or the vehicle cannot continue), the vehicle reports it to the system. The system immediately notifies the system operator, who takes care of the incident. The incident is stored in the vehicle's history together with its routes and route changes for troubleshooting purposes.
	Primary Actors	Vehicle
	Secondary Actors	System operator
	Preconditions	
	Main Flow of Events	
	Alternative Flows	

	Post-conditions	

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Entity Relationship Diagram

Entity Relationship Diagram1



Summary

Name	Description
Incident	
Employee	
Person	
Waypoint	
Route	
RegisteredUser	
PremiumMembership	
Order_Passenger	
ParkingSpot	
Vehicle	
Order	
PaymentCard	
Payment	
ChargingStation	
Vehicle_Feature	
Order_Feature	
Feature	

N/A	The start of a route is the waypoint with the lowest sequence_number, the end is the one with the highest. Each route has at least 2 waypoints.

Details

Incident
Employee
Person
Waypoint
Route
RegisteredUser
PremiumMembership
Order_Passenger
ParkingSpot
Vehicle
Order
PaymentCard
Payment
ChargingStation

Vehicle_Feature
Order_Feature
Feature
N/A